



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2005
HIGHER SCHOOL CERTIFICATE
TRIAL PAPER

Mathematics Extension 2

General Instructions

- Reading Time – 5 Minutes
- Working time – 3 Hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Section is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 120

- Attempt questions 1 – 8

Examiner: *C.Kourtesis*

NOTE: This is a trial paper only and does not necessarily reflect the content or format of the final Higher School Certificate examination paper for this subject.

Section A
(Start a new answer sheet.)

Question 1. (15 marks)

Marks

(a) Evaluate $\int_0^2 \frac{3}{4+x^2} dx$. **2**

(b) Find $\int \cos x \sin^4 x \, dx$. **1**

(c) Use integration by parts to find $\int te^{-t} dt$. **2**

(d) (i) Find real numbers a and b such that **2**

$$\frac{1}{x(\pi-2x)} = \frac{a}{x} + \frac{b}{\pi-2x}.$$

(ii) Hence find **2**

$$\int \frac{dx}{x(\pi-2x)}.$$

(e) Evaluate $\int_{-3}^3 (2-|x|) dx$. **2**

(f) (i) Use the substitution $x = a - t$ to prove that **2**

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx.$$

(ii) Hence evaluate **2**

$$\int_0^{\frac{\pi}{2}} \log_e (\tan x) dx$$

Question 2. (15 marks)

Marks

- (a) If $z = 2 + i$ and $w = -1 + 2i$ find

2

$$\operatorname{Im}(z - w).$$

- (b) On an Argand diagram shade the region that is satisfied by both the conditions

2

$$\operatorname{Re}(z) \geq 2 \text{ and } |z - 1| \leq 2.$$

- (c) If $|z| = 2$ and $\arg z = \theta$ determine

3

(i) $\left| \frac{i}{z^2} \right|$ (ii) $\arg\left(\frac{i}{z^2}\right)$

- (d) If for a complex number z it is given that $\bar{z} = z$ where $z \neq 0$, determine the locus of z .

2

- (e) A complex number z is such that $\arg(z + 2) = \frac{\pi}{6}$ and $\arg(z - 2) = \frac{2\pi}{3}$.

3

Find z , expressing your answer in the form $a + ib$ where a and b are real.

- (f) The complex numbers z_1 , z_2 and z_3 are represented in the complex plane by the points P , Q and R respectively. If the line segments PQ and PR have the same length and are perpendicular to one another, prove that:

3

$$2z_1^2 + z_2^2 + z_3^2 = 2z_1(z_2 + z_3)$$

Section B
(Start a new answer sheet.)

Question 3. (15 marks)

Marks

- | | | |
|------|---|----------|
| (a) | If $2 - 3i$ is a zero of the polynomial $z^3 + pz + q$ where p and q are real, find the values of p and q . | 3 |
| (b) | If α , β and γ are roots of the equation $x^3 + 6x + 1 = 0$ find the polynomial equation whose roots are $\alpha\beta$, $\beta\gamma$ and $\alpha\gamma$. | 2 |
| (c) | Consider the function $f(x) = 3\left(\frac{x+4}{x}\right)^2$. | |
| (i) | Show that the curve $y = f(x)$ has a minimum turning point at $x = -4$ and a point of inflexion at $x = -6$. | 5 |
| (ii) | Sketch the graph of $y = f(x)$ showing clearly the equations of any asymptotes. | 2 |
| (d) | Use mathematical induction to prove that | 3 |
| | $n! > 2^n$ for $n > 3$ where n is an integer. | |

Question 4 (15 marks)

(a) If $f(x) = \sin x$ for $-\pi \leq x \leq \pi$ draw neat sketches, on separate diagrams, of:

(i) $y = [f(x)]^2$ 2

(ii) $y = \frac{1}{f\left(x + \frac{\pi}{2}\right)}$ 2

(iii) $y^2 = f(x)$ 2

(iv) $y = f(\sqrt{|x|})$ 2

(b) Show that the equation of the tangent to the curve $x^{\frac{1}{2}} + y^{\frac{1}{2}} = a^{\frac{1}{2}}$ at the point $P(x_0, y_0)$ on the curve is $xx_0^{-\frac{1}{2}} + yy_0^{-\frac{1}{2}} = a^{\frac{1}{2}}$. 3

(c) Consider the polynomial $P(x) = x^5 - ax + 1$. By considering turning points on the curve $y = P(x)$, prove that $P(x) = 0$ has three distinct roots if $a > 5\left(\frac{1}{2}\right)^{\frac{8}{5}}$. 4

Section C
(Start a new answer booklet)

Question 5 (15 marks)

Marks

- (a) A particle of mass m is thrown vertically upward from the origin with initial speed V_0 . The particle is subject to a resistance equal to mkv , where v is its speed and k is a positive constant.

- (i) Show that until the particle reaches its highest point the equation of motion is

1

$$\ddot{y} = -(kv + g)$$

where y is its height and g is the acceleration due to gravity.

- (ii) Prove that the particle reaches its greatest height in time T given by

4

$$kT = \log_e \left[1 + \frac{kV_0}{g} \right].$$

- (iv) If the highest point reached is at a height H above the ground prove that

4

$$V_0 = Hk + gT.$$

- (b) If α and β are roots of the equation $z^2 - 2z + 2 = 0$

- (i) find α and β in mod-arg form.

3

- (ii) show that $\alpha^n + \beta^n = \sqrt{2^{n+2}} \cdot \left[\cos \frac{n\pi}{4} \right].$

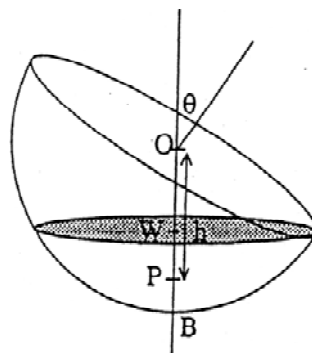
3

Question 6 (15 marks)

- (a) A group of 20 people is to be seated at a long rectangular table, 10 on each side. There are 7 people who wish to sit on one side of the table and 6 people who wish to sit on the other side. How many seating arrangements are possible? 2

- (b) The area enclosed by the curves $y = \sqrt{x}$ and $y = x^2$ is rotated about the y axis through one complete revolution. Use the cylindrical shell method to find the volume of the solid that is generated. 3

- (c) The diagram shows a hemi-spherical bowl of radius r . The bowl has been tilted so that its axis is no longer vertical, but at an angle θ to the vertical. At this angle it can hold a volume V of water.



The vertical line from the centre O meets the surface of the water at W and meets the bottom of the bowl at B . Let P be between W and B , and let h be the distance OP .

- (i) Explain why $V = \int_{r \sin \theta}^r \pi(r^2 - h^2)dh$. 3
- (ii) Hence show $V = \frac{r^3 \pi}{3} (2 - 3 \sin \theta + \sin^3 \theta)$. 2
- (d) (i) Show that $x^4 + y^4 \geq 2x^2y^2$. 2
- (ii) If $P(x,y)$ is any point on the curve $x^4 + y^4 = 1$ prove that $OP \leq 2^{\frac{1}{4}}$, where O is the origin. 3

Section D
(Start a new answer booklet)

Question 7 (15 marks)

- (a) How many sets of 5 quartets (groups of four musicians) can be formed from 5 violinists, 5 viola players, 5 cellists, and 5 pianists if each quartet is to consist of one player of each instrument? 2

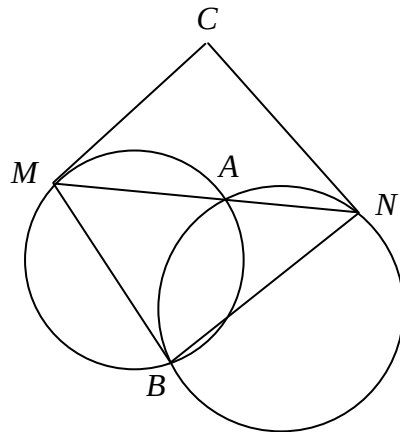
- (b) (i) If $t = \tan \theta$, prove that 2

$$\tan 4\theta = \frac{4t(1-t^2)}{1-6t^2+t^4}.$$

- (ii) If $\tan \theta \tan 4\theta = 1$ deduce that $5t^4 - 10t^2 + 1 = 0$. 2

- (iii) Given that $\theta = \frac{\pi}{10}$ and $\theta = \frac{3\pi}{10}$ are roots of the equation 4
 $\tan \theta \tan 4\theta = 1$, find the exact value of $\tan \frac{\pi}{10}$.

(c)

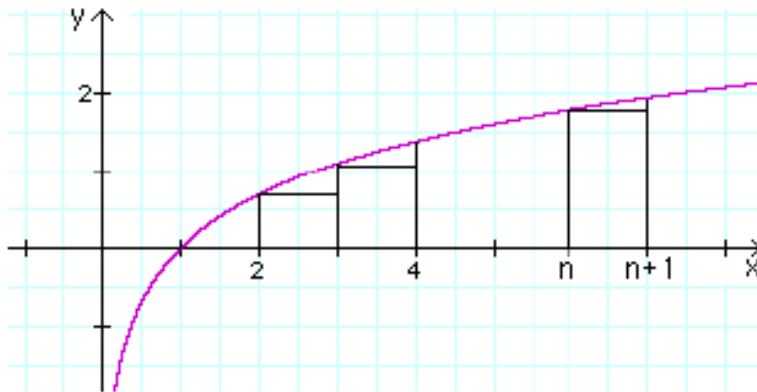


Two circles intersect at A and B . A line through A cuts the circles at M and N . The tangents at M and N intersect at C . 5

- (i) Prove that $\angle CMA + \angle CNA = \angle MBN$.
(ii) Prove M, C, N, B are concyclic.

Question 8 (15 marks)

(a)



6

The diagram above shows the graph of $y = \log_e x$ for $1 \leq x \leq n+1$.

- (i) By considering the sum of the areas of inner and outer rectangles show that

$$\ln(n!) < \int_1^{n+1} \ln x \, dx < \ln[(n+1)!]$$

- (ii) Find $\int_1^{n+1} \ln x \, dx$.

- (iii) Hence prove that

$$e^n > \frac{(n+1)^n}{n!}$$

- (b) If a root of the cubic equation $x^3 + bx^2 + cx + d = 0$ is equal to the reciprocal of another root, prove that

3

$$1 + bd = c + d^2.$$

This question continues on the next page.

- (c) A stone is projected from a point O on a horizontal plane at an angle of elevation α and with initial velocity U metres per second. The stone reaches a point A in its trajectory, and at that instant it is moving in a direction perpendicular to the angle of projection with speed V metres per second. **6**

Air resistance is neglected throughout the motion and g is the acceleration due to gravity.

If t is the time in seconds at any instant, show that when the stone is at A :

(i) $V = U \cot \alpha$

(ii) $t = \frac{U}{g \sin \alpha}.$

This is the end of the paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

$$\text{NOTE: } \ln x = \log_e x, x > 0$$



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AUGUST 2005

**Trial Higher School Certificate
Examination**

YEAR 12

Mathematics Extension 2

Sample Solutions

Section	Marker
A	PP
B	EC
C	PB
D	DH

Section A

$$\begin{aligned}
 (1) \quad (i) \quad \int_0^2 \frac{3}{4+x^2} dx &= \frac{3}{2} \int_0^2 \frac{3}{4+x^2} dx \\
 &= \frac{3}{2} \left[\tan^{-1} \left(\frac{x}{2} \right) \right]_0^2 \\
 &= \frac{3}{2} [\tan^{-1}(1) - 0] \\
 &= \frac{3\pi}{8}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \int \cos x \sin^4 x dx &= \int u^4 du \quad [u = \sin x] \\
 &= \frac{u^5}{5} + c \\
 &= \frac{\sin^5 x}{5} + c
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad \int \underbrace{t}_{g} \underbrace{e^{-t}}_{f'} dt &= fg - \int fg' dt \\
 &= -te^{-t} - \int (-e^{-t} \times 1) dt \\
 &= -te^{-t} + \int e^{-t} dt \\
 &= -te^{-t} - e^{-t} + c
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad (i) \quad 1 &\equiv a(\pi - 2x) + bx \\
 x = 0 &\Rightarrow a = \frac{1}{\pi} \\
 2a &= b \text{ [coefficients of } x] \\
 \therefore b &= \frac{2}{\pi}
 \end{aligned}$$

$$a = \frac{1}{\pi}, b = \frac{2}{\pi}$$

$$\begin{aligned}
 (ii) \quad \int \frac{dx}{x(\pi - 2x)} &= \frac{1}{\pi} \int \left(\frac{1}{x} - \frac{-2}{\pi - 2x} \right) dx \\
 &= \frac{1}{\pi} \ln x - \frac{1}{\pi} \ln(\pi - 2x) + c \\
 &= \frac{1}{\pi} \ln \left(\frac{x}{\pi - 2x} \right) + c
 \end{aligned}$$

$$\begin{aligned}
 \text{(e)} \quad \int_{-3}^3 (2 - |x|) dx &= 2 \int_0^3 (2 - |x|) dx && \left[\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ if } f \text{ is even} \right] \\
 &= 2 \int_0^3 (2 - x) dx && \left[\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ if } f \text{ is even} \right] \\
 &= 2 \left[2x - \frac{1}{2} x^2 \right]_0^3 \\
 &= 2 \left[6 - \frac{9}{2} \right] \\
 &= 3
 \end{aligned}$$

$$\begin{aligned}
 \text{(f)} \quad \text{(i)} \quad x &= a - t \Rightarrow dx = -dt \\
 x &= 0 \Rightarrow t = a \\
 x &= a \Rightarrow t = 0 \\
 \int_0^a f(x) dx &= \int_a^0 f(a-t) (-dt) \\
 &= \int_0^a f(a-t) dt && \left[\int_a^b f(x) dx = - \int_b^a f(x) dx \right] \\
 &= \int_0^a f(a-x) dx && \left[\text{Q choice of variable irrelevant} \right]
 \end{aligned}$$

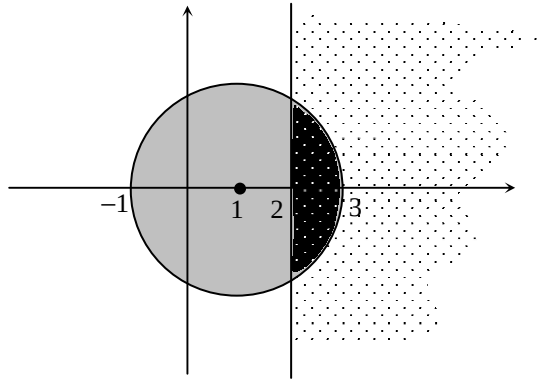
$$\begin{aligned}
 \text{(ii)} \quad I &= \int_0^{\frac{\pi}{2}} \ln(\tan x) dx \\
 &= \int_0^{\frac{\pi}{2}} \ln \left(\tan \left(\frac{\pi}{2} - x \right) \right) dx \\
 &= \int_0^{\frac{\pi}{2}} \ln(\cot x) dx \\
 2I &= \int_0^{\frac{\pi}{2}} \ln(\tan x) dx + \int_0^{\frac{\pi}{2}} \ln(\cot x) dx \\
 &= \int_0^{\frac{\pi}{2}} \left[\ln(\tan x) + \ln(\cot x) \right] dx \\
 &= \int_0^{\frac{\pi}{2}} \ln 1 dx \\
 &= 0 \\
 \therefore 2I &= 0 \\
 \therefore I &= 0 \\
 \therefore \int_0^{\frac{\pi}{2}} \ln(\tan x) dx &= 0
 \end{aligned}$$

(2) (a) $z = 2 + i, w = -1 + 2i$

$$\therefore z - w = 3 - i$$

$$\therefore \operatorname{Im}(z - w) = -1$$

(b)



(c) (i) $\left| \frac{i}{z^2} \right| = \frac{|i|}{|z|^2} = \frac{1}{4}$

(ii) $\arg\left(\frac{i}{z^2}\right) = \arg i - \arg(z^2)$

$$= \frac{\pi}{2} - 2 \arg z$$

$$= \frac{\pi}{2} - 2\theta$$

(d) $z = \bar{z} \Rightarrow z$ is purely real

So the locus is $y = 0$, except $x = 0$.

Alternatively:

Let $z = x + iy, (z \neq 0)$

$$\therefore \bar{z} = x - iy$$

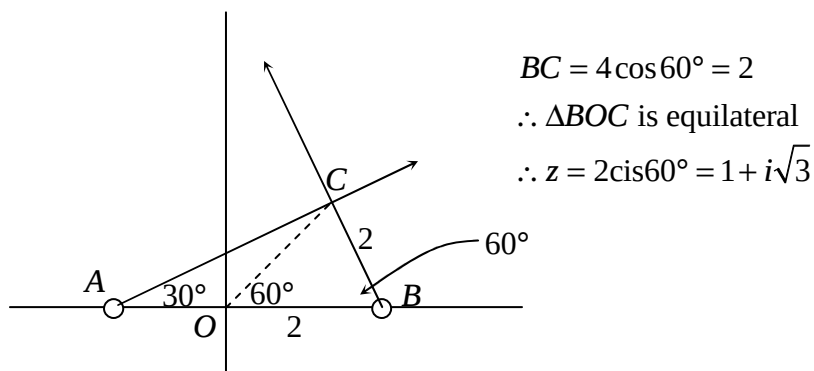
$$\therefore z = \bar{z} \Rightarrow x + iy = x - iy$$

$$\therefore 2iy = 0 \Rightarrow y = 0$$

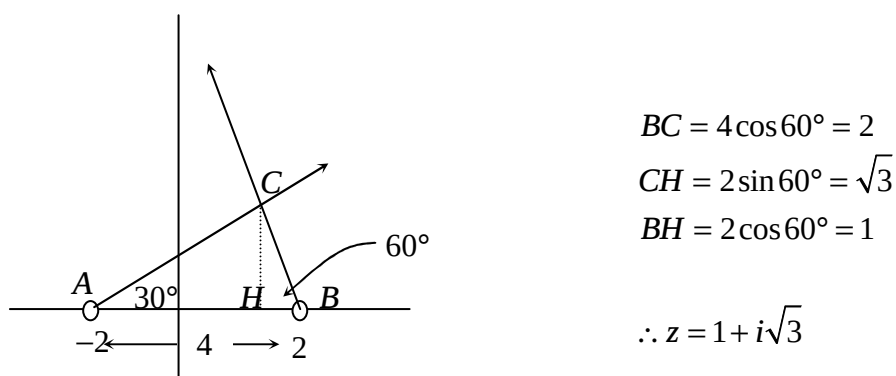
$\therefore z$ is a purely real number excluding 0

(e) $\arg(z+2) = \frac{\pi}{6}, \arg(z-2) = \frac{2\pi}{3}.$

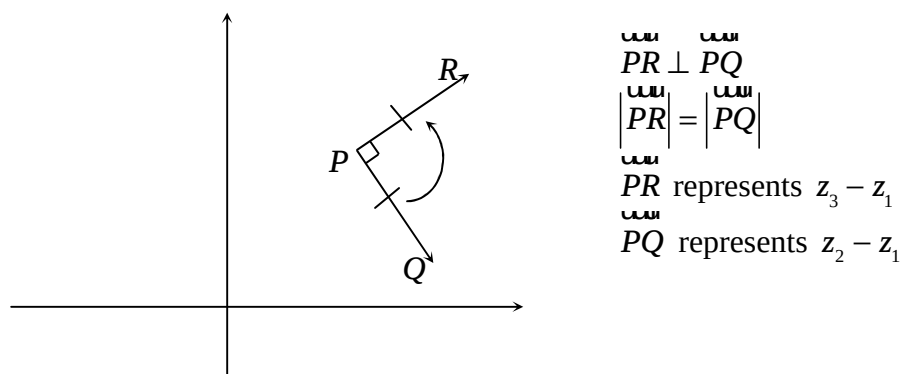
z is represented by the point C , the intersection of the two rays.



Alternatively



(f)



$$i(z_2 - z_1) = z_3 - z_1$$

$$\therefore i^2 (z_2 - z_1)^2 = (z_3 - z_1)^2$$

$$\therefore -(z_2^2 - 2z_2z_1 + z_1^2) = z_3^2 - 2z_1z_3 + z_1^2$$

$$\therefore 2z_1^2 + z_2^2 + z_3^2 = 2z_1z_3 + 2z_2z_1 = 2z_1(z_2 + z_3)$$

Section B

(a) $z^3 + pz + q = 0$
 If $(2-3i)$ is a zero then
 $(2-3i)^3 + (2-3i)p + q = 0$
 $(2-3i)^3 = (2-3i)^2(2-3i)$
 $= (-5-12i)(2-3i)$
 $= -46-9i$
 $-46-9i + (2-3i)p + q = 0$
 Equating real parts.
 $-46 + 2p + q = 0$
 $2p + q = 46$ — (1)
 Equating imaginary parts.
 $-9 - 3p = 0 \Rightarrow 3p = -9$
 $p = -3$ — (2)
 Subst. (2) into (1)
 $-6 + q = 46$
 $q = 52$ — (3)
[3]

Question (3)

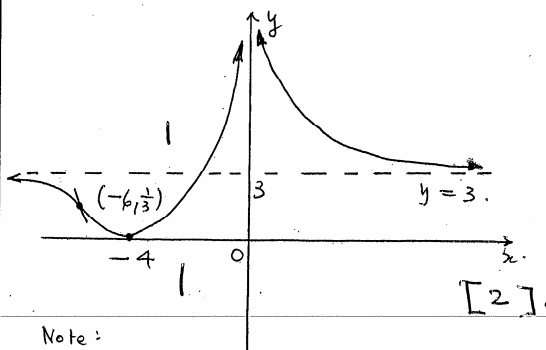
(b) $x^3 + 6x + 1 = 0$
 If α, β, γ are the roots
 then $\alpha\beta\gamma = -1$
 Now, $\alpha\beta = \frac{\alpha\beta\gamma}{\gamma} = -\frac{1}{\gamma}$
 Similarly, $\beta\gamma = -\frac{1}{\alpha}$
 and $\alpha\gamma = -\frac{1}{\beta}$
 $\therefore$ Let $y = -\frac{1}{x} \Rightarrow x = -\frac{1}{y}$
 i.e. the polynomial equation
 is $(-\frac{1}{y})^3 + 6(-\frac{1}{y}) + 1 = 0$
 $-1 - 6y^2 + y^3 = 0$
 $y^3 - 6y^2 - 1 = 0$ **[2]**

(c) $y = 3\left(\frac{x+4}{x}\right)^2$
 $\frac{dy}{dx} = 6\left(\frac{x+4}{x}\right)\left(-\frac{4}{x^2}\right)$
 $= -\frac{24(x+4)}{x^3}$
 $\frac{dy}{dx} = 0 \Rightarrow x = -4$ 2
 When $x = -4$, $y = 0$
 $\therefore (-4, 0)$ is a stationary pt.
 $\frac{d^2y}{dx^2} = -24\left[\frac{x^3 - (x+4)3x^2}{x^6}\right]$
 $= 24\left[\frac{2x^3 + 12x^2}{x^6}\right]$
 $= 48x^2\left(\frac{x+6}{x^4}\right)$
 $f''(x) = 0, \Rightarrow x = -6$
 When $x = -6$, $y = 3\left(\frac{-6+4}{-6}\right)^2$
 $= \frac{1}{3}$ 2
 $\therefore (-6, \frac{1}{3})$ could be a
 pt. of inflexion.

Test:

x	-7	-6	-5
$f(x)$	-	0	+

$\therefore (-6, \frac{1}{3})$ is a pt. of inflexion.
 Also, $f''(-4) = \frac{48}{-4^4}(2) > 0$
 $\therefore (-4, 0)$ is a min turning pt.
 $x \neq 0 \therefore y$ -axis is a vertical
 asymptote. **[5]**



Note:

$\lim_{x \rightarrow \pm\infty} 3\left(\frac{x^2+8x+16}{x^2}\right) = 3$
 i.e. horizontal asymptote at $y = 3$.

(d) Let $S(n)$ be the proposition

that $n! > 2^n$ $n > 3$.

$n \in \mathbb{Z}^+$.

For $n=4$

$$4! = 24 > 2^4 = 16$$

$\therefore S(4)$ is true

Assume $S(k)$ is true.

Prove true for $n=k+1$.

$$(k+1)! = (k+1)k!$$

$$> (k+1) \cdot 2^k$$

$\therefore k > 3$, then $k+1 > 4$

$$> 4 \cdot 2^k$$

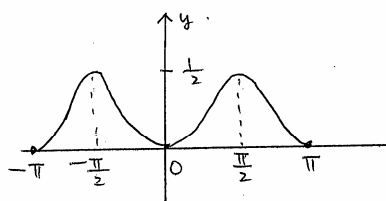
$$2 \cdot 2^{k+1}$$

$$> 2^{k+1}$$

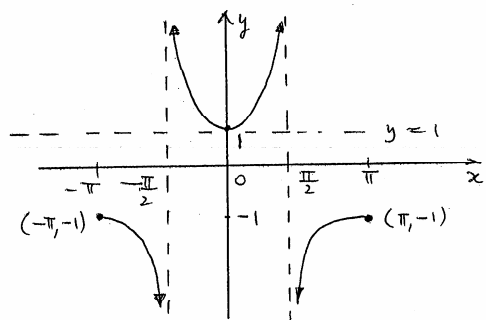
Question (4).

$$y = \sin x, \quad -\pi \leq x \leq \pi$$

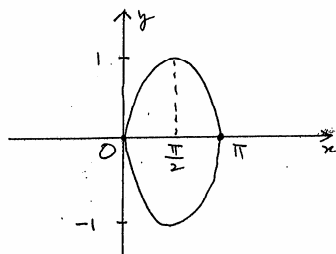
$$(i) \quad y = \sin^2 x \\ = \frac{1}{2}(1 - \cos 2x)$$



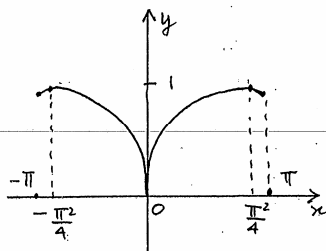
$$(ii) \quad y = \frac{1}{\sin(x+\pi/2)} = \frac{1}{\cos x}$$



$$(iii) \quad y^2 = \sin x \\ y = \pm \sqrt{\sin x}$$



(iv)



$$(b) \quad x^{1/2} + y^{1/2} = a^{1/2} \\ \therefore \frac{1}{2}x^{-1/2} + \frac{1}{2}y^{-1/2} \cdot \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{x^{1/2}}{y^{1/2}} = -\sqrt{\frac{y}{x}}$$

$$\frac{dy}{dx} \Big|_{x=x_0, y=y_0} = -\sqrt{\frac{y_0}{x_0}}$$

Equation of tangent.

$$y - y_0 = -\frac{y_0^{1/2}}{x_0^{1/2}}(x - x_0)$$

$$\therefore x_0^{1/2}y - x_0^{1/2}y_0 = x_0y_0^{1/2} - xy_0^{1/2}$$

$$\Rightarrow y_0^{1/2}x + x_0^{1/2}y = x_0^{1/2}y_0^{1/2}(y_0^{1/2} + x_0^{1/2})$$

$\therefore (x_0, y_0)$ is on the curve — (1)

$$\Rightarrow x_0^{1/2} + y_0^{1/2} = a^{1/2}$$

and divide both sides of (1)

by $x_0^{1/2}y_0^{1/2}$ we have.

$$x_0^{-1/2}x + y_0^{-1/2}y = a^{1/2}$$

$$\begin{aligned}
 (c) \quad p(x) &= x^5 - ax + 1 \\
 p'(x) &= 5x^4 - a \\
 p''(x) &= 20x^3
 \end{aligned}$$

Stationary pts when

$$5x^4 - a = 0$$

$$x^4 = \frac{a}{5}$$

$$\therefore x = \pm \left(\frac{a}{5}\right)^{\frac{1}{4}}$$

$$\text{For } x = \left(\frac{a}{5}\right)^{\frac{1}{4}}, p''\left[\left(\frac{a}{5}\right)^{\frac{1}{4}}\right] > 0$$

$$\text{For } x = -\left(\frac{a}{5}\right)^{\frac{1}{4}}, p''\left[-\left(\frac{a}{5}\right)^{\frac{1}{4}}\right] < 0$$

$$\text{When } x = \left(\frac{a}{5}\right)^{\frac{1}{4}}, y = \left(\frac{a}{5}\right)^{\frac{5}{4}} - a\left(\frac{a}{5}\right)^{\frac{1}{4}} + 1$$

$$\therefore y = 1 - \left(\frac{4a}{5}\right)\left(\frac{a}{5}\right)^{\frac{1}{5}}$$

$$\text{For } x = -\left(\frac{a}{5}\right)^{\frac{1}{4}}$$

$$y = 1 + \left(\frac{4a}{5}\right)\left(\frac{a}{5}\right)^{\frac{1}{5}}$$

For three distinct real roots, the curve cuts the x-axis in 3 points.

$$x \rightarrow \infty \quad p(x) \rightarrow \infty (>0)$$

and the turning points are on the opposite sides of the x-axis

$\Rightarrow$ The product of the y's < 0

$$\therefore \left[1 + \left(\frac{4a}{5}\right)\left(\frac{a}{5}\right)^{\frac{1}{5}}\right] \left[1 - \left(\frac{4a}{5}\right)\left(\frac{a}{5}\right)^{\frac{1}{5}}\right] < 0$$

$$1 - \frac{16a^2}{25}\left(\frac{a}{5}\right)^{\frac{1}{5}} < 0$$

$$\frac{16a^{5/2}}{5^{5/2}} > 1 \Rightarrow a^{5/2} > \left(\frac{5^{5/2}}{2^4}\right)$$

$$\therefore a > \left(\frac{5^{5/2}}{2^4}\right)^{2/5} = \left[\frac{5^1}{2^{8/5}}\right]$$

$$\Rightarrow a > 5\left(\frac{1}{2}\right)^{\frac{8}{5}} \quad [4]$$

Section C

QUESTIONS

$$(a) (i) \quad m\ddot{y} = -mkv - mg \quad \begin{array}{ccc} \uparrow & \downarrow & \downarrow \\ m\ddot{y} & mg & mkv \end{array}$$

$$\therefore \boxed{\ddot{y} = -(kv + g)}$$

$$(ii) \quad \ddot{y} = \frac{dv}{dt} = -(kv + g)$$

$$\therefore dt = -(kv + g)^{-1} dv$$

$$\int_0^T 1 \cdot dt = - \int_{v_0}^0 (kv + g)^{-1} dv$$

$$T = \int_0^{v_0} (kv + g)^{-1} dv$$

$$= \int_0^{v_0} \frac{1}{kv + g} dv$$

$$= \frac{1}{k} \left[\ln(kv + g) \right]_0^{v_0}$$

$$= \frac{1}{k} \left(\ln(kv_0 + g) - \ln g \right)$$

$$= \frac{1}{k} \ln \left(\frac{kv_0 + g}{g} \right)$$

$$\therefore \boxed{kT = \ln \left(1 + \frac{kv_0}{g} \right)}$$

$$(11) \quad v \cdot \frac{dv}{dy} = -(kv + g)$$

$$\therefore \frac{dv}{dy} = -\left(\frac{kv + g}{v}\right)$$

$$dy = \frac{-v \cdot dv}{kv + g}$$

$$= -\frac{1}{k} \left(\frac{kv + g - g}{kv + g} \right) dv$$

$$\int_0^H 1 \cdot dy = -\frac{1}{k} \int_{V_0}^0 \frac{kv + g - g}{kv + g} dv$$

$$\therefore H = \frac{1}{k} \int_0^{V_0} \left(1 - \frac{g}{kv + g} \right) dv$$

$$= \frac{1}{k} \left[\int_0^{V_0} 1 \cdot dv - \int_0^{V_0} \frac{g}{kv + g} dv \right]$$

$$= \frac{1}{k} \left[[v]_0^{V_0} - \left[\frac{g}{k} \ln(kv + g) \right]_0^{V_0} \right]$$

$$= \frac{1}{k} \left[V_0 - \frac{g}{k} (\ln(kv_0 + g) - \ln g) \right]$$

$$kH = V_0 - \frac{g}{k} \ln \frac{kv_0 + g}{g}$$

$$= V_0 - \frac{g}{k} kT \quad \text{from (1)}$$

$$= V_0 - gT$$

$$\therefore \boxed{V_0 = kH + gT}$$

$$\begin{aligned}
 (b) \quad (i) \quad z &= \frac{2 \pm \sqrt{4-8}}{2} \\
 &= \frac{2 \pm 2i}{2} \\
 &= 1 \pm i \\
 &= \sqrt{2} \operatorname{cis} \pm \frac{\pi}{4}
 \end{aligned}$$

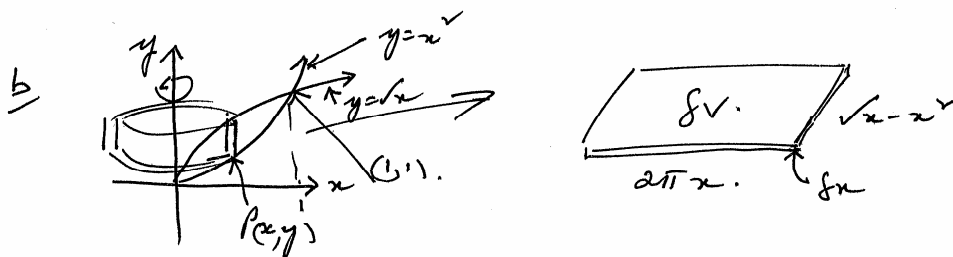
$$\therefore \alpha, \beta \text{ are } \sqrt{2} \operatorname{cis} \pm \frac{\pi}{4}$$

$$\begin{aligned}
 (ii) \quad \alpha^n + \beta^n &= \left(\sqrt{2} \operatorname{cis} \frac{\pi}{4} \right)^n + \left(\sqrt{2} \operatorname{cis} -\frac{\pi}{4} \right)^n \\
 &= (\sqrt{2})^n \left[\operatorname{cis} \frac{n\pi}{4} + \operatorname{cis} -\frac{n\pi}{4} \right] \\
 &= (\sqrt{2})^n \left(2 \cos \frac{n\pi}{4} \right) \quad \left(\begin{array}{l} \text{NB} \\ z + \bar{z} = 2 \operatorname{Re} z \end{array} \right) \\
 &= 2^{\frac{1}{2}n+1} \cos \frac{n\pi}{4} \\
 &= 2^{\frac{n+2}{2}} \cos \frac{n\pi}{4}
 \end{aligned}$$

$$\therefore \boxed{\alpha^n + \beta^n = \sqrt{2^{n+2}} \cos \frac{n\pi}{4}}$$

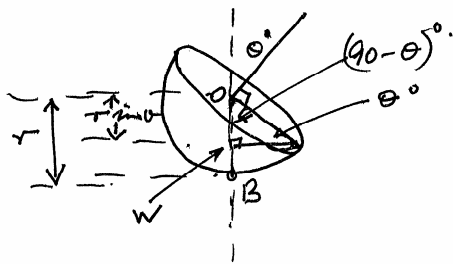
QUESTION 6.

a $\boxed{{}^{10}P_7 \times {}^{10}P_6 \times 7!}$ (this is the same as $\binom{10}{7} \times \binom{10}{6} \times (7!)^2 \times 6!$)



$$\begin{aligned}
 V &= \lim_{\delta x \rightarrow 0} \sum_{x=0}^1 2\pi x (\sqrt{x - x^2}) \delta x \\
 &= 2\pi \int_0^1 (x^{3/2} - x^3) dx \\
 &= 2\pi \cdot \left[\frac{2}{5} x^{5/2} - \frac{x^4}{4} \right]_0^1 \\
 &= 2\pi \left(\frac{2}{5} - \frac{1}{4} \right) \\
 &= 2\pi \times \frac{3}{20} \\
 &= \boxed{\frac{3\pi}{10}}
 \end{aligned}$$

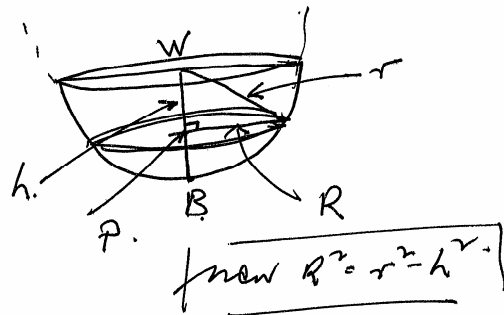
(c) (1)



$$\text{NB } \frac{OW}{r} = \sin \theta$$

$$|OW = r \sin \theta|$$

$$\text{and } |OB = r|$$



$$\text{now } R^2 = r^2 - h^2$$

Consider the slice at P.



$$\delta V = \pi R^2 \delta h.$$

$$\therefore V = \lim_{\delta h \rightarrow 0} \sum_{h=r \sin \theta}^{h=r} \pi R^2 \delta h.$$

$$= \pi \int_{r \sin \theta}^r R^2 \cdot dh.$$

$$= \pi \int_{r \sin \theta}^r (r^2 - h^2) \cdot dh.$$

$$(11) \quad V = \pi \left[r^2 h - \frac{h^3}{3} \right]_{r \sin \theta}^r$$

$$= \pi \left[r^3 - \frac{r^3}{3} - \left(r^3 \sin \theta - \frac{r^3 \sin^3 \theta}{3} \right) \right]$$

$$= \pi \left[\frac{2r^3}{3} - r^3 \sin \theta + \frac{r^3 \sin^3 \theta}{3} \right]$$

$$\left[= \frac{\pi r^3}{3} (2 - 3 \sin \theta + \sin^3 \theta) \right]$$

$$(d) (i) \text{ now } (x^r - y^r)^2 \geq 0.$$

$$x^4 - 2x^r y^r + y^4 \geq 0$$

$$\therefore \underline{x^4 + y^4 \geq 2x^r y^r}$$

$$(ii) \text{ now } OP = \sqrt{x^r + y^r}$$

$$\therefore OP^4 = (x^r + y^r)^2$$

$$= x^4 + y^4 + 2x^r y^r$$

$$\leq x^4 + y^4 + x^4 + y^4 \text{ (from (i))}$$

$$\leq 2. \quad (x^4 + y^4 = 1)$$

$$\therefore \underline{OP \leq 2^{\frac{1}{4}}}$$

Section D

7. (a) How many sets of 5 quartets (groups of four musicians) can be formed from 5 violinists, 5 viola players, 5 cellists, and 5 pianists if each quartet is to consist of one player of each instrument?

Solution: $\frac{5^4 \times 4^4 \times 3^4 \times 2^4 \times 1^4}{5!} = (5!)^3,$ $= 1\,728\,000.$
--

- (b) i. If $t = \tan \theta$, prove that

$$\tan 4\theta = \frac{4t(1-t^2)}{1-6t^2+t^4}.$$

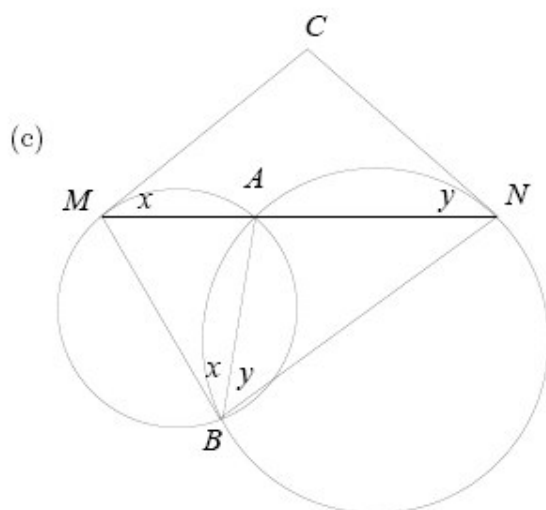
Solution: L.H.S. $= \frac{2 \times \tan 2\theta}{1 - (\tan 2\theta)^2},$ $= \frac{2 \times \frac{2t}{1-t^2}}{1 - \left(\frac{2t}{1-t^2}\right)^2},$ $= \frac{4t(1-t^2)}{(1-t^2)^2 - 4t^2},$ $= \frac{4t(1-t^2)}{1-6t^2+t^4},$ $= \text{R.H.S.}$
--

- ii. If $\tan \theta \tan 4\theta = 1$, deduce that $5t^4 - 10t^2 + 1 = 0$.

Solution: $t \times \frac{4t(1-t^2)}{1-6t^2+t^4} = 1,$ $4t^2 - 4t^4 = 1 - 6t^2 + t^4,$ $5t^4 - 10t^2 + 1 = 0.$

- iii. Given that $\theta = \frac{\pi}{10}$ and $\theta = \frac{3\pi}{10}$ are roots of the equation $\tan \theta \tan 4\theta = 1$, find the exact value of $\tan \frac{\pi}{10}$.

Solution: Using the quadratic formula, $t^2 = \frac{10 \pm \sqrt{100-20}}{5},$ $= \frac{5 \pm 2\sqrt{5}}{5}.$ $i.e., t = \sqrt{\frac{5 \pm \sqrt{5}}{5}} \text{ as } \tan \frac{\pi}{10}, \tan \frac{3\pi}{10} > 0.$ Now, as $\tan \frac{\pi}{10} < \tan \frac{3\pi}{10}, \tan \frac{\pi}{10} = \sqrt{\frac{5 - \sqrt{5}}{5}}.$
--



Two circles intersect at A and B . A line through A cuts the circles at M and N . The tangents at M and N intersect at C .

- i. Prove that $\angle CMA + \angle CNA = \angle MBN$.

Solution: Join AB .

$$\angle CMA = \angle MBA \text{ (angle in alternate segment),}$$

$$\angle CNA = \angle ABN \text{ (angle in alternate segment),}$$

$$\begin{aligned} \therefore \angle CMA + \angle CNA &= \angle MBA + \angle ABN, \\ &= \angle MBN. \end{aligned}$$

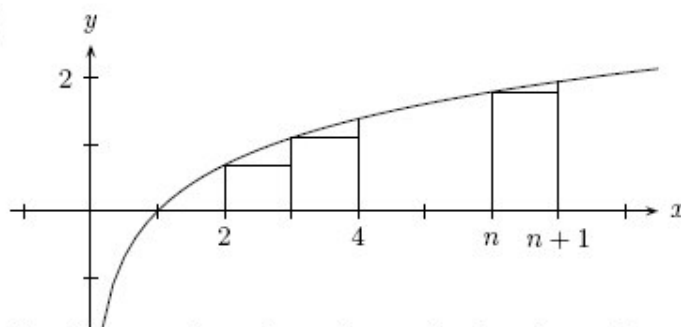
- ii. Prove M, C, N, B are concyclic.

Solution: $\angle CMA + \angle CNA + \angle MCN = 180^\circ$ (angle sum of $\triangle CMN$),

$$\therefore \angle MBN + \angle MCN = 180^\circ.$$

So $MCNB$ is a cyclic quadrilateral (opposite angles supplementary).

8. (a)



The diagram above shows the graph of $y = \log_e x$ for $1 \leq x \leq n+1$.

- i. By considering the sum of the areas of inner and outer rectangles, show that

$$\ln(n!) < \int_1^{n+1} \ln x \, dx < \ln((n+1)!)$$

$$\begin{aligned}
\text{Solution: Sum inner rectangles} &= \sum_{x=1}^n \ln x \times 1, \\
&= \ln 1 + \ln 2 + \ln 3 + \cdots + \ln n, \\
&= \ln n! \\
\text{Sum outer rectangles} &= \sum_{x=2}^{n+1} \ln x \times 1, \text{ or } \sum_{x=1}^n \ln(x+1) \times 1, \\
&= \ln 2 + \ln 3 + \ln 4 + \cdots + \ln(n+1), \\
&= \ln(n+1)! \\
\therefore \ln n! &< \int_1^{n+1} \ln x \, dx < \ln(n+1)!
\end{aligned}$$

ii. Find $\int_1^{n+1} \ln x \, dx$.

$$\begin{aligned}
\text{Solution: } I &= \int_1^{n+1} \ln x \times 1 \, dx, & u &= \ln x & v' &= 1 \\
&= [x \ln x]_1^{n+1} - \int_1^{n+1} dx, & u' &= \frac{1}{x} & v &= x \\
&= (n+1) \ln(n+1) - 0 - [x]_1^{n+1}, \\
&= (n+1) \ln(n+1) - (n+1-1), \\
&= (n+1) \ln(n+1) - n.
\end{aligned}$$

iii. Hence prove that

$$e^n > \frac{(n+1)^n}{n!}$$

$$\begin{aligned}
\text{Solution: From i., } \ln(n+1)! &> \int_1^{n+1} \ln x \, dx. \\
\therefore \ln(n+1)! &> (n+1) \ln(n+1) - n. \\
n &> \ln \frac{(n+1)^{n+1}}{(n+1)!}, \\
&> \ln \frac{(n+1)^n}{n!}. \\
\therefore e^n &> \frac{(n+1)^n}{n!}.
\end{aligned}$$

(b) If a root of the cubic equation $x^3 + bx^2 + cx + d = 0$ is equal to the reciprocal of another root, prove that

$$1 + bd = c + d^2.$$

Solution: Let the roots be $\alpha, \frac{1}{\alpha}, \beta$.

Method 1:

$$\begin{aligned}
\alpha \times \frac{1}{\alpha} \times \beta &= -d, \\
\beta &= -d.
\end{aligned}$$

Substitute in the equation for the root β :

$$\begin{aligned}
 -d^3 + bd^2 - cd + d &= 0, \\
 cd + d^3 &= bd^2 + d. \\
 \text{Divide by } d \ (d \neq 0), \\
 c + d^2 &= bd + 1.
 \end{aligned}$$

Method 2:

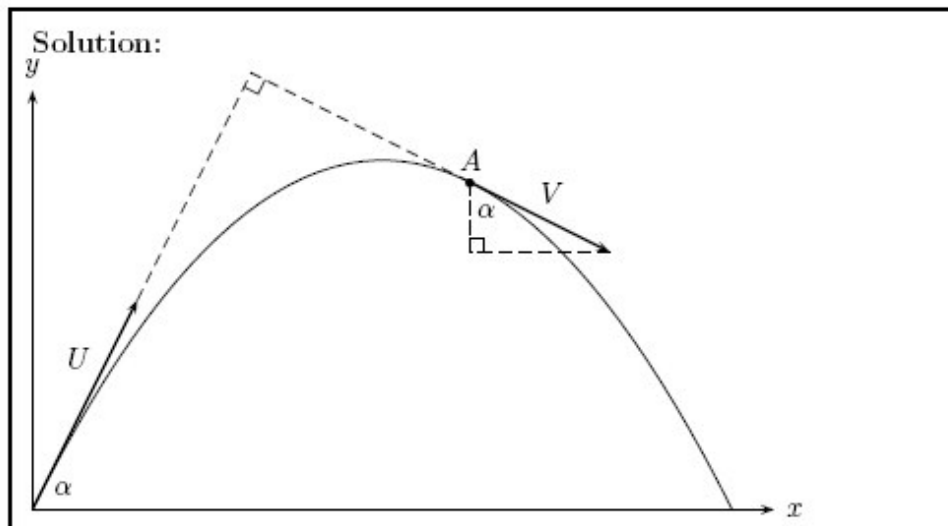
$$\begin{aligned}
 \alpha + \frac{1}{\alpha} + \beta &= -b, \\
 1 + \alpha\beta + \frac{\beta}{\alpha} &= c, \\
 \beta &= -d. \\
 \therefore \alpha + \frac{1}{\alpha} - d &= -b \dots \boxed{1} \\
 1 - \alpha d - \frac{d}{\alpha} &= c \dots \boxed{2} \\
 1 - c &= d\left(\alpha + \frac{1}{\alpha}\right), \\
 \therefore \alpha + \frac{1}{\alpha} &= \frac{1-c}{d}. \\
 \text{Sub. in } \boxed{1}, \frac{1-c}{d} - d &= -b, \\
 1 - c - d^2 &= -bd, \\
 \text{i.e., } 1 + bd &= c + d^2.
 \end{aligned}$$

- (c) A stone is projected from a point O on a horizontal plane at an angle of elevation α and with initial velocity U metres per second. The stone reaches a point A in its trajectory, and at that instant it is moving in a direction perpendicular to the angle of projection with speed V metres per second.

Air resistance is neglected throughout the motion and g is the acceleration due to gravity.

If t is the time in seconds at any instant, show that when the stone is at A :

- i. $V = U \cot \alpha$



$$\begin{aligned}\ddot{x} &= 0 & \ddot{y} &= -g \\ \dot{x} &= U \cos \alpha & \dot{y} &= U \sin \alpha - gt \\ \text{At } A, \quad U \cos \alpha &= V \sin \alpha, \\ & \text{i.e., } V &= U \cot \alpha\end{aligned}$$

ii. $t = \frac{U}{g \sin \alpha}$

Solution: At A , $\dot{y} = -V \cos \alpha$ (now heading downwards),
i.e., $-U \cot \alpha \times \cos \alpha = U \sin \alpha - gt$,

$$\begin{aligned}gt &= U \sin \alpha + U \frac{\cos \alpha}{\sin \alpha} \cdot \cos \alpha, \\ &= U \left(\frac{\sin^2 \alpha + \cos^2 \alpha}{\sin \alpha} \right). \\ \therefore t &= \frac{U}{g \sin \alpha}.\end{aligned}$$